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Graded Assignment • 30 min

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Hallucinations in LLMs

LLM use cases: Chatbots, Academic uses

Risks real and imagined

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- **Reading:** Required: Scientific American article on AI risks
  10 min
- **Reading:** Optional: AI-generated e-books and journalism
  30 min
- **Reading:** Optional: Generative AI, copyright, and Hollywood strikes
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Module 3 Quiz

Assignment details

Due
Dec 24, 11:59 PM IST

Attempts
Unlimited

Submitted
Dec 3, 5:02 AM IST

Your grade
To pass you need at least 80%. We keep your latest score.

93.33%

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Your latest: **93.33%** • Your highest: **93.33%** • To pass you need at least 80%. We keep your latest score.

1. What is the main concern about automated AI detectors expressed in the linked Stanford blog post? **1 / 1 point**

- ( ) They cannot identify content written by native English speakers.
- ( ) They are too accurate in identifying AI-generated content.
- (●) They are unreliable, especially for non-native English writers.
- ( ) They are easily fooled by non-native English speakers.

Correct
The main concern is the unreliability of AI detectors, particularly when dealing with non-native English writers, as highlighted in the article.

2. Which of the following is **NOT** an issue with the use of automated AI detectors in an academic context? **1 / 1 point**

- ( ) Current AI detectors have been shown to disproportionally detect writing from non-native speakers.
- ( ) Current AI detectors are not reliable, producing both false positives and false negatives.
- ( ) The use of AI is not necessarily academic misconduct in all contexts, and detecting its use should not necessarily be penalised.
- (●) AI detectors are expensive and inefficient.

Correct
Reduced efficiency in identifying potential instances of plagiarism is not a typical risk associated with using automated AI detectors for academic writing. In fact, the use of AI detectors is generally intended to improve efficiency.

3. Which is the main current mitigation strategy for hallucinations in LLMs? **1 / 1 point**

- ( ) Ensuring diverse and representative training data.
- (●) Human-in-the-loop verification.
- ( ) Regular fine-tuning of the model.
- ( ) Adversarial testing to identify and correct hallucinatory outputs.

Correct

4. In the Bender and Hanna article from Scientific American, what do the authors consider to be the real threat from artificial intelligence? **1 / 1 point**

- ( ) The lack of understanding about AI's capabilities
- (●) The spread of misinformation through AI-generated content
- ( ) Existential risks leading to the end of humanity
- ( ) The potential misuse of AI in warfare

Correct
The authors emphasise that the real threat from artificial intelligence lies in existing dangers, such as misinformation and harm caused by AI-generated content.

5. What is the main goal of the Stanford One Hundred Year Study on Artificial Intelligence? **0 / 1 point**

- ( ) To develop advanced AI models for various industries.
- (●) To predict the future impact of AI on society.
- ( ) To create a comprehensive AI curriculum for universities.
- ( ) To study and anticipate the long-term effects of AI on humanity.

Incorrect

6. True or False: The authors of the Bender and Hanna article believes that AI technology can effectively address gaps in education and healthcare for those in need. **1 / 1 point**

- ( ) True
- (●) False

Correct
The article argues that the deployment of AI technology can actually hurt workers and points out the negative impact on various sectors.

7. What is the correct definition of **prompt injection**? **1 / 1 point**

- ( ) Prompt injection is a technique used to secure machine learning models by restricting access to prompts.
- ( ) Prompt injection refers to the process of generating creative prompts for artistic projects.
- ( ) Prompt injection involves replacing human-generated prompts with computer-generated instructions to enhance model performance.
- (●) Prompt injection is a family of exploits where malicious users manipulate a model's behavior by feeding it crafted instructions.

Correct
Prompt injection is a family of exploits where malicious users manipulate a machine learning model's behavior by providing it with crafted instructions, often replacing the intended human-generated prompts. This can lead to security vulnerabilities and undesirable behavior in the model.

8. What is a crucial distinction between human hallucinations and what is often called "hallucinations" in generative AI? **1 / 1 point**

- ( ) AI hallucinations are deliberate attempts at creativity, while human hallucinations are unintentional.
- (●) AI hallucinations arise from incorrect data interpretation, while human hallucinations arise from incorrect sensory experiences.
- ( ) AI hallucinations are based on rational thought processes, while human hallucinations are random.
- ( ) AI hallucinations are always sensory experiences, while human hallucinations can be conceptual.

Correct
A crucial distinction between "hallucinations" in AI and human hallucinations is that AI hallucinations arise from incorrect data interpretation or lack of context, whereas human hallucinations are typically sensory experiences perceived by individuals and may not necessarily be accurate representations of reality.

9. What was the model from OpenAI that preceded ChatGPT? **1 / 1 point**

- ( ) Codex
- (●) InstructGPT
- ( ) GPT-3.5
- ( ) Azure AI

Correct
ChatGPT builds on the development of InstructGPT

10. What is an example of a sector where hallucinations in AI-generated content could have serious consequences? **0 / 1 point**

- (●) Social media and personal communication.
- ( ) Music composition and performance.
- ( ) Entertainment and creative writing.
- ( ) Weather forecasting and climate modeling.

Incorrect

11. What emotional and psychological impact have annotators reported while preparing training data for Large Language Models such as GPT? **1 / 1 point**

- ( ) Heightened sense of emotional detachment from the content being annotated.
- ( ) Enhanced emotional resilience through constant exposure to sensitive content.
- ( ) Increased creativity and job satisfaction from exposure to diverse content.
- (●) Stress, anxiety, and discomfort due to exposure to distressing or offensive material.

Correct
Annotators have reported stress, anxiety, and discomfort when exposed to distressing or offensive content during the annotation process.

12. What is one of the allegations made in the lawsuits filed in the USA against OpenAI (GPT) and Meta (LLaMA) regarding generative AI and copyright? **1 / 1 point**

- ( ) The models directly replicate copyrighted works without alteration.
- (●) The models remix copyrighted works without consent, compensation, or credit.
- ( ) The models only use openly available public domain content for training.
- ( ) The models generate entirely original content not based on copyrighted materials.

Correct
One of the allegations made in the lawsuits is that the models, both from OpenAI and Meta, remix copyrighted works without obtaining consent, compensation, or giving credit to the original authors. This raises concerns about potential copyright infringement.

13. Which of the following is NOT a task that would be carried out by human judges when training or fine-tuning a language model. **1 / 1 point**

- ( ) Creating the initial dataset for training.
- (●) Designing the architecture of the AI model.
- ( ) Generating reward signals for reinforcement learning.
- ( ) Providing feedback on model-generated responses.

Correct
Human judges are not responsible for designing the architecture of the model.

14. What is the distinction between task-based and social chatbots? **1 / 1 point**

- ( ) Task-based chatbots excel in complex decision-making, while social chatbots specialize in casual conversations.
- ( ) Task-based chatbots are designed for entertainment, while social chatbots are intended for professional use.
- (●) Task-based chatbots are programmed to accomplish specific tasks, while social chatbots aim to engage in open-ended and human-like conversations.
- ( ) Task-based chatbots prioritize providing emotional support, while social chatbots focus on completing specific tasks.

Correct
The distinction between task-based and social chatbots is that task-based chatbots are programmed to accomplish specific tasks, such as setting reminders or making reservations, while social chatbots focus on engaging in open-ended and human-like conversations, often for the purpose of entertainment or companionship.

15. What was the primary focus of the open letter issued by the Center for AI Safety in May 2023? **1 / 1 point**

- ( ) Promoting AI as a solution to societal-scale challenges.
- (●) Addressing the potential risk of extinction from AI as a global priority.
- ( ) Highlighting the economic potential of AI technologies.
- ( ) Advocating for the immediate halt of all AI research and development.

Correct
The open letter from the Center for AI Safety emphasized the importance of recognizing and addressing the risk of extinction from AI as a significant global concern.

16. How did the Stanford study find that AI detectors performed in classifying TOEFL essays written by non-native English students? **1 / 1 point**

- ( ) They were highly accurate in detecting human-written essays
- ( ) They correctly identified most of the essays as AI-generated
- (●) They were unreliable, classifying over half as AI-generated
- ( ) They performed better than in evaluating U.S.-born eighth-graders' essays

Correct
The study revealed that the AI detectors were unreliable when classifying TOEFL essays written by non-native English students. They mistakenly identified more than half of these essays as AI-generated.

17. As of July 2023, what is the typical advice given to students at UK universities about the use of AI tools such as GPT? **1 / 1 point**

- (●) Students should learn how to use AI tools effectively and acknowledge their use appropriately.
- ( ) Students should use AI tools freely to generate assessment content without limitations.
- ( ) Students should avoid using AI tools altogether to ensure academic integrity.
- ( ) Students should rely solely on AI tools for all assessment content to improve efficiency.

Correct

18. Why is a pretrained LLM like GPT not suitable for use in an interactive chatbot in its default configuration? **1 / 1 point**

- ( ) Off-the-shelf LLMs can only generate text and cannot engage in speech-based interactions.
- (●) Off-the-shelf LLMs generate pre-trained responses that might not align with the desired conversation.
- ( ) Off-the-shelf LLMs lack the ability to understand human language.
- ( ) Off-the-shelf LLMs generate responses that are too short and lack context.

Correct
An off-the-shelf LLM generates pre-trained responses that might not align with the desired conversation or the context of the interaction. Fine-tuning or customizing the LLM's responses based on specific requirements is necessary to make it useful in an interactive context such as a chatbot.

19. Which of the following is **NOT** a recommended prompt engineering technique for students? **1 / 1 point**

- (●) Giving as little information as possible to allow for more creative output.
- ( ) Requesting sources for all pieces of generated information.
- ( ) Providing specific information about the desired length and target audience.
- ( ) Using iterative queries and asking follow-up questions.

Correct

20. What contributes to ChatGPT's ability to simulate creative output? **1 / 1 point**

- ( ) Its capacity to generate content without any reference to patterns.
- (●) Its training on a diverse range of human-written content.
- ( ) Its access to personal experiences and emotions.
- ( ) Its deep understanding of consciousness and self-awareness.

Correct
ChatGPT's ability to simulate creative output is based on its training on various human-written content, which enables it to mimic language patterns and generate responses that appear creative within the scope of its training.

21. Current AI detectors typically base their scores on **perplexity**, which correlates with the sophistication of the writing. What is the main issue with this design choice? **1 / 1 point**

- ( ) Perplexity measures the complexity of the AI's internal structure.
- ( ) Perplexity is not a reliable measure of language fluency.
- ( ) Perplexity does not consider the context of the text.
- (●) Perplexity can lead to biases against non-native English writers.

Correct
The main issue with using perplexity as the main metric for AI detectors is that it can introduce biases against non-native English writers or individuals with less sophisticated writing skills. This is because perplexity scores might be lower for these writers, leading to incorrect classifications as AI-generated content.

22. What aspect of generative AI is often a focus of regulatory efforts? **1 / 1 point**

- ( ) Ensuring AI models have no access to internet data.
- (●) Ensuring transparency and explainability of AI systems.
- ( ) Ensuring that AI-generated content is always original.
- ( ) Ensuring that AI models are completely predictable.

Correct
Regulatory efforts often focus on ensuring transparency and explainability of AI systems, which helps build trust and accountability.

23. The chatbot deployed by the American National Eating Disorders Association (NEDA) was disabled shortly after it went live in June 2023. What was the main reason for this? **1 / 1 point**

- (●) It was providing harmful eating disorder advice to users.
- ( ) It was generating irrelevant responses to user queries.
- ( ) It was causing technical glitches in NEDA's helpline system.
- ( ) It was too expensive to maintain and operate.

Correct
The NEDA chatbot was disabled in June 2023 because it was providing harmful eating disorder advice to users, prompting the organization to take action.

24. What is the main distinction between ChatGPT and InstructGPT? **1 / 1 point**

- (●) ChatGPT is designed for generating conversational responses, while InstructGPT is optimized for generating instructional content.
- ( ) ChatGPT is specialized in generating programming code, while InstructGPT focuses on creative writing.
- ( ) ChatGPT is exclusively trained on medical terminology, while InstructGPT is trained on legal language.
- ( ) ChatGPT is designed for translating text between different languages, while InstructGPT focuses on speech recognition.

Correct
The main difference lies in the primary use cases of the two models: ChatGPT is intended for generating natural language conversation, while InstructGPT is designed for providing detailed responses based on instructions.

25. Why is it not appropriate to use an LLM such as GPT as a search engine directly? **1 / 1 point**

- ( ) LLMs generate long and detailed responses, making it difficult to find relevant information quickly.
- ( ) LLMs are too slow to retrieve search results in a timely manner.
- ( ) LLMs have limited language comprehension and cannot understand search queries.
- (●) LLMs do not in general have access to live, real-time data, which means that any response may not necessarily be accurate.

Correct

26. Which of the following is NOT an AI harm identified by Bender and Hanna? **1 / 1 point**

- ( ) Routine discrimination in housing, criminal justice, and health care.
- ( ) Defamation and deep-fake pornography.
- (●) Existential risks and potential to wipe out humanity.
- ( ) Spread of hate speech and misinformation in non-English languages.

Correct

27. What is a significant limitation of ChatGPT that is mentioned in the OpenAI blog post introducing it? **1 / 1 point**

- ( ) Its over-reliance on the InstructGPT dataset
- ( ) Its inability to respond to any questions
- ( ) Its inability to provide verbose responses
- (●) Writing plausible-sounding but incorrect answers

Correct
The blog post mentions that ChatGPT sometimes writes plausible-sounding but incorrect answers.

28. What should teachers and lecturers avoid when using GPT for academic purposes? **1 / 1 point**

- ( ) Creating marking criteria and rubrics for assignments.
- ( ) Using GPT to brainstorm ideas for assignment questions.
- (●) Writing from scratch without any reference material.
- ( ) Editing existing academic documents with GPT.

Correct

29. What consequence did lawyers in New York face for submitting briefs that contained content generated by ChatGPT? **1 / 1 point**

- ( ) They were commended for their innovative use of technology in the legal field.
- (●) They were each fined for submitting misleading information.
- ( ) They were rewarded with a grant for their use of AI-generated content.
- ( ) They received recognition for contributing to AI research and development.

Correct
The lawyers in New York were fined $5000 each for submitting briefs that cited non-existent court decisions generated by ChatGPT. This was due to the misleading nature of the information they presented in the case.

30. Following the reported incident in March 2023, which is the policy now adopted by the literary magazine Clarkesworld? **1 / 1 point**

- ( ) They accept submissions written with AI tools but with additional review.
- ( ) They accept all submissions regardless of the tools used.
- ( ) They actively encourage authors to use AI writing tools for their submissions.
- (●) They do not consider submissions written, developed, or assisted by AI tools.

Correct
Clarkesworld's policy is that they will not consider submissions written, developed, or assisted by AI writing tools like ChatGPT. Attempting to submit such works may result in being banned from submitting in the future.